Application Components Overview

This section describes the major application components of the Automated Fraud Detection System, organized according to the Layered Architecture style. Each layer has a specific responsibility and interacts only with its adjacent layers, improving modularity, maintainability, and scalability.

1. Presentation Components

- **Customer Interface:** Displays transaction status and fraud alerts to users.
- **Fraud Analyst Dashboard:** Enables analysts to view flagged transactions and investigate fraud cases.
- **System Admin Console:** Allows administrators to manage rules, AI models, and compliance settings.

2. Application / Service Components

- **Transaction Monitoring Service:** Monitors real-time transactions and behavioral patterns.
- **Risk Scoring Service:** Computes fraud risk scores for each transaction.
- **Transaction Decision Service:** Approves, flags, or blocks transactions based on risk thresholds.
- **Manual Review Service:** Sends suspicious transactions for human investigation.
- **Alert Management Service:** Generates and dispatches fraud alerts in real time.

3. Business Logic Components

- **Fraud Detection Rules Engine:** Applies predefined and dynamic fraud detection rules.
- **AI / ML Risk Evaluation Engine:** Uses machine learning models to identify anomalies and predict fraud.
- **Risk Classification Module:** Classifies transactions into Low, Medium, or High risk categories.

4. Integration Components

- **External Fraud Database Connector:** Interfaces with external blacklists and fraud intelligence sources.
- **Notification Service Adapter:** Integrates with SMS, Email, and Push notification systems.
- **Banking / Payment Gateway Interface:** Communicates with core banking and payment processing systems.

5. Data Management Components

- **Transaction Database:** Stores transaction details, risk scores, and outcomes.
- **User and Role Database:** Maintains user profiles, roles, and access permissions.
- **Audit and Compliance Repository:** Stores logs and audit trails for regulatory compliance.
- **Case Management Database:** Records fraud investigation history and analyst actions.

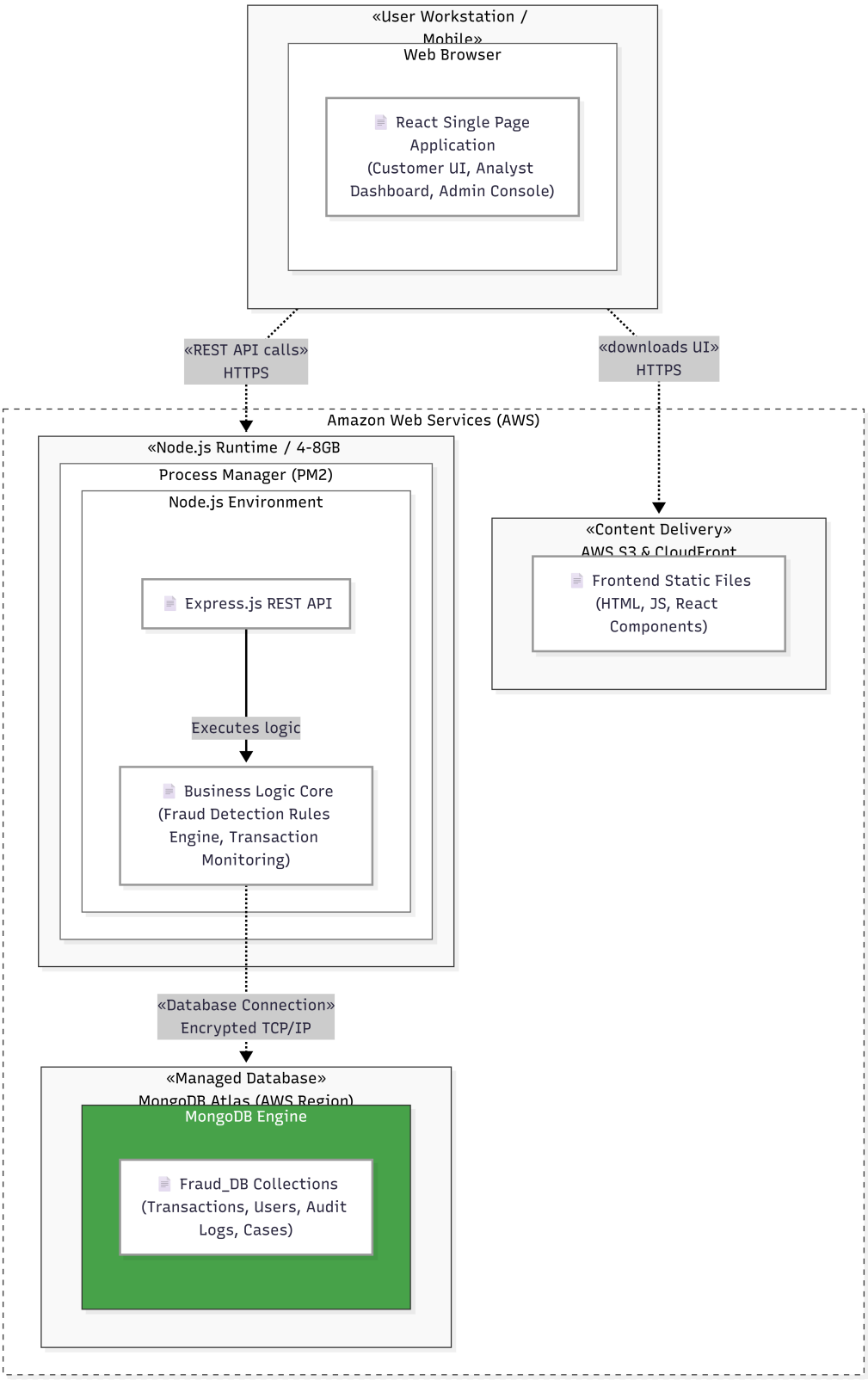

Figure 1: Deployment Architecture for the Automated Fraud Detection System

I. Hosting and Deployment Strategy

Host Site

The deployment of the Automated Fraud Detection System will leverage AWS cloud services to ensure high availability and scalability across the layered MERN stack architecture.

- **Presentation Components:** The Customer Interface, Fraud Analyst Dashboard, and System Admin Console will be built as a React application. These frontend components will be hosted on an AWS S3 bucket configured for static website hosting and distributed globally via AWS CloudFront.
- **Application/Service and Business Logic Components:** These backend layers, encompassing the Transaction Monitoring Service, Fraud Detection Rules Engine, and integration connectors, will run on a Node.js/Express environment. This backend will be deployed on AWS EC2 instances or AWS Elastic Beanstalk.
- **Data Management Components:** The Transaction Database, User and Role Database, Audit and Compliance Repository, and Case Management Database will utilize MongoDB. These will be hosted on MongoDB Atlas, deployed within the same AWS region as the backend servers to minimize latency.

Deployment Strategy

1. **Database Provisioning:** Initialize the MongoDB Atlas clusters to store all transaction details, risk scores, logs, and investigation history. Secure the clusters by whitelisting only the IP addresses of the backend AWS servers.
2. **Backend Server Setup:** Deploy the Express and Node.js application codebase to AWS. Configure the server environment securely by injecting environment variables (such as the MongoDB URI and API keys) using AWS Systems Manager Parameter Store.
3. **API Configuration:** Expose the backend services via an AWS API Gateway. Configure Cross-Origin Resource Sharing (CORS) on the Express server to strictly accept incoming API requests only from the production domain of the S3-hosted React frontend.
4. **Frontend Deployment:** Generate the optimized production build of the React application. Upload these static build files to the AWS S3 bucket and invalidate the CloudFront cache so users receive the most up-to-date application interfaces.

Security Measures

- **Data Encryption:** Enforce HTTPS/SSL using AWS Certificate Manager for all data in transit across the Presentation, Application, and Data Management layers. Ensure encryption at rest is enabled on all MongoDB Atlas instances to protect sensitive logs and user profiles.
- **Network Firewalls:** Use AWS Virtual Private Cloud (VPC) to deploy the backend resources in private subnets. Implement AWS Security Groups to restrict traffic, ensuring the backend only accepts traffic from the API Gateway and only makes authorized outbound connections to MongoDB and the Banking Gateway.
- **Access Control:** Implement robust JWT (JSON Web Token) authentication to protect the endpoints utilized by the Fraud Analyst Dashboard and System Admin Console, validating permissions against the User and Role Database.